Updated Individual Assignment Report

SI649-18FALL-Xi Li

Executive Summary

I will create a visualization to help explain permutation and combination, 2 related mathematical concepts which requires a strong “abstracting” ability without a distinctive method to “display” the problem. Reference will be:
http://mathworld.wolfram.com/Permutation.html for permutation, which asks for a consideration for order issue.
http://mathworld.wolfram.com/Combination.html for combination, which does not talk about order issue.

The idea of it is to use visualizing the items in problem to help students understand and differentiate them better and solve problems with the correct method.
Here I will provide both a visualizing introduction together with explanation and math formulas as well as exercise for student to get a better understanding (visual part is more for the beginner status, while the exercise is more for advanced).

This is for people from advanced high school or above (pre-college).

Learning Objectives (1 page or less)

3-5 learning objectives for my project.

1. The student/viewer will be able to identify the difference between permutation and combination.
2. The student/viewer will be able to define which method to be used in a specific context / problem given.
3. The student/viewer will be able to practice with combination/permutation problems by themselves through inputting key numbers and check the outputs, to check their learnings (see open answer question below). This helps them understand better about abstract problems.

Exam

1. Multiple choice: There are 3 balls in a invisible bag, 1 blue, 1 red and 1 yellow. You pick one out for each time and put it back to bag after checking the color. In a total of 3 times' picking, which one of the following events has a probability of 8/9?
    - Same color for all 3 times
    - Totally different colors for all 3 times
    - Not all of the the colors are the same
    - No red color
2. Open answer: In a natural reserve, the population of pandas inside is N. One day, researchers have T pandas to check their health conditions. Special marks are made for them before having them back to the natural reserve. After a year, supposing that the population of panda does not change and M pandas are checked for health conditions this time. Among M pandas, what is the probably that S out of M pandas have marks from the 1st time?

My screenshots

Due to that a permutation of inputting total number over than 4 can generate a pretty long list of visual elements and as there are too many items included, I decided to not visualize every input but to separate visualizing part (which is more from a communication way) and knowledge part. And users can still finish the “Permutation/Combination” learning through a step-by-step way. Meanwhile, they will be asked to input some numbers to generate results which can provide an interactive and engaging feeling for them to get a better understanding (as well as many times of trying as they can change the numbers).

Combination & Permutation

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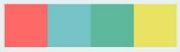
To get started, I want to show you what is the difference between these 2 terms visually first.

Please choose an interger below and check what's happening:

2 3 4

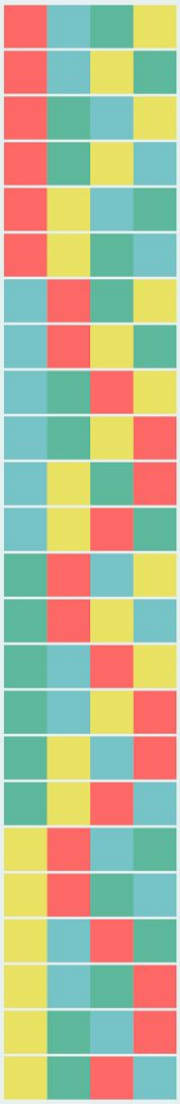


Great! Now you see there are some squares. If I ask you to select some of them to me, either in order or not, how many are you willing to give me?

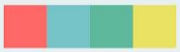
1 2 3 4

RESET

Permutation



Combination



From "play with squares", you can see that the major difference between combination and permutation is if there is an order issue.

Definition

What is Combination?

$$C_k^n = \binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

The number of ways of picking k unordered outcomes from n possibilities.

What is Permutation?

$$P_k^n = \frac{n!}{(n-k)!}$$

A permutation, also called an "arrangement number" or "order," is a rearrangement of the elements of an ordered list S into a one-to-one correspondence with S itself. When you are trying to select k items out of n, permutation includes all possible ordered possibilities.

please input the total amount of squares you want in a bag as "n":

5

please input how many squares you want to pick out from the bag as "k":

2

SUBMIT RESET

Combination Result:

10

Permutation Result:

20

Reference

http://mathworld.wolfram.com/Permutation.html

http://mathworld.wolfram.com/Combination.html

http://pebreo.github.io/combinations-visualization/

*color palette and design style: inspired by styles of ecoach used in umich 1st year course like STATS250

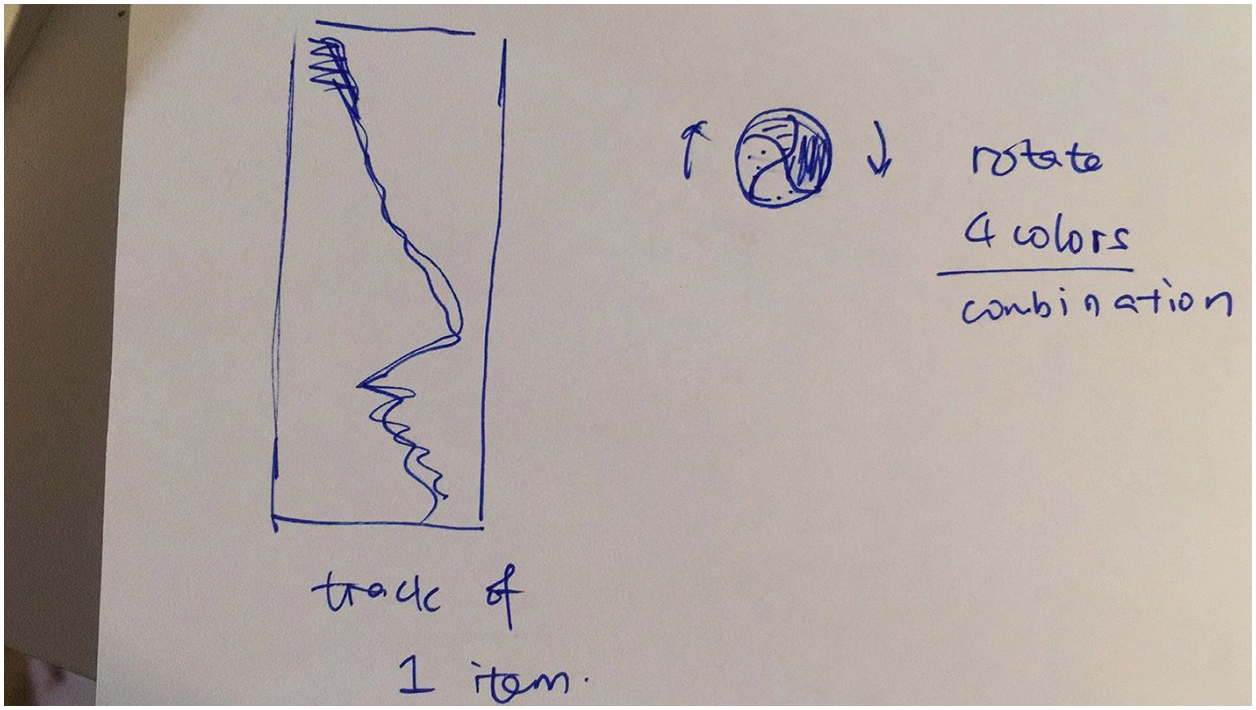
Why will your vis help?

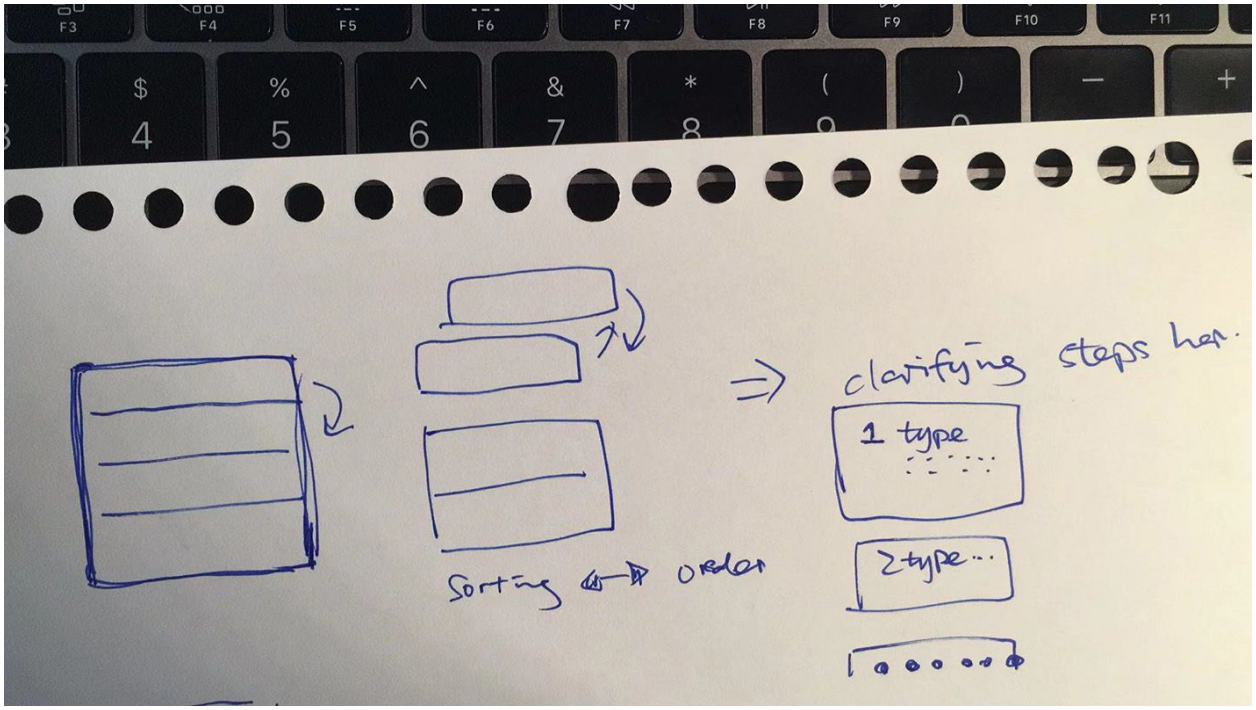
One challenge in understanding and differentiating permutation & combination, and use the function will be how to list all possibilities.

From my past experience, when I am new to these concepts, I always need to draw items in different ways on sketching paper to help me kind of get to know the concept better in order to solve a question. Therefore, an interactive tool will assist well since I can check if I calculate in the correct way based on comparison between my “abstract” calculated answer and the visual elements shown on the screen.

A limitation of my solution is that there is still limited freedom in playing with it. And due to that there were still some problems with my previous js code (I finished a color generating code and had it into use to generate an array of n colors which represented the total input of items before. But got some problem in making permutations algorithm. So I use color palette datasets directly instead, to show users the possible outputs as you can see in current webpage). So there were only limited visualizing results. It could be more interesting if users could see the visualizing outputs as they input numbers n and k into practicing boxes under Definition. Apart from that, it is still a more traditional presenting, so if I could make something like “drag and drop squares into answer box and check if your answers are correct” could be more interactive.

Archive of Experiments

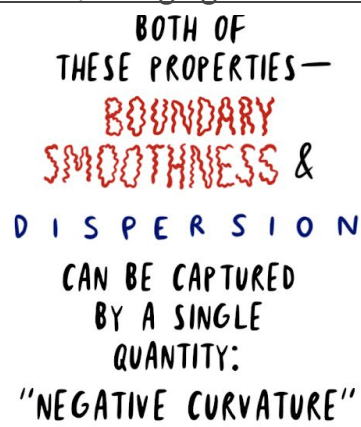

rotate
4 colors
combination
track of
1 item.

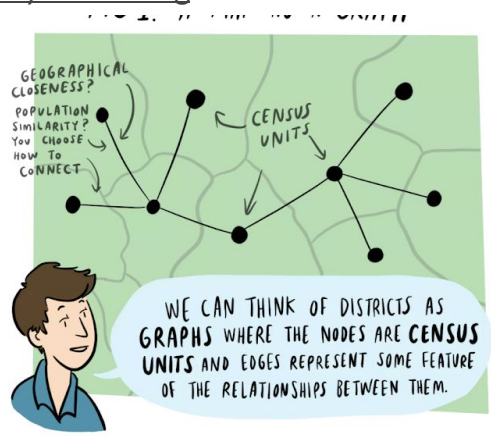

clarifying steps here.
1 type
2 type...
Sorting
order

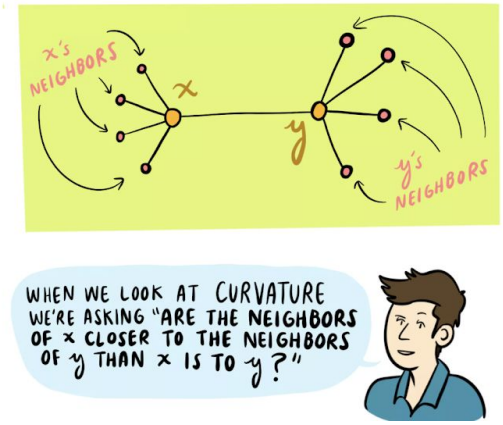
Inspiration/Competitive Analysis

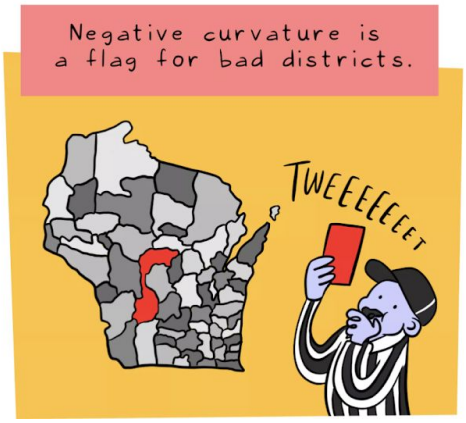
Some inspirations and competitor as well I got are from the online resources:

https://thenib.com/changing-the-math-on-gerrymandering









A github project about permutation and combination concept:
http://pebreo.github.io/combinations-visualization/

Peer-Robin Note

Previously, I am thinking of making something for climate change. But the scope goes too large and even I tried to limit it down, there are still some data problems appeared during the time I made it. So actually I took the feedbacks from last time and thinking of changing to another topic which may be clear and have more interactive inputs to teach students something instead of more about giving them information.